Redbird Flight Simulations

Qualification and Approval Guide (QAG)

TD BASIC- Basic Aviation Training Device
Amended Qualification and Approval Guide
VERSION 2.0
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Section 1: List of Configurations

Currently Approved:

1. **Configuration: TD BASIC-Standard -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

2. **Configuration: TD BASIC-Glass -** Single Engine, Land, Fixed Gear, Fixed Pitch Propeller, High Wing

Statement of Compatibility of Software and Hardware
AC61-136, Appendix 2, 4. b.:

The devices and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.
“This is to certify that Microsoft Corporation, the owner and developer of the ESP and FSX Simulator Engine, has evaluated that their software application works with industry standard PC’s and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.”

Section 2: Description of Hardware and Software Components

All Configurations Include:

Qty	Manufacturer	Configuration	Description/Function
1	Redbird Flight Simulations	All	Central Processor– Sim Platform Software
1	Microsoft	All	Operating System
1	Microsoft	All	Simulation Engine (FSX or ESP depending on serial number)
1	Redbird Flight Simulations	All	Analogy Instruments, Hobbs, Flap and Trim Indicators
1	Mindstar Aviation	See Configuration	Redbird 530
1	Mindstar Aviation	See Configuration	Redbird 140 Autopilot
1	Mindstar Aviation	See Configuration	Redbird 1000
1	Flight 1	All	Compass, Miscellaneous Gauges
1	Microsoft	All	Map, Track, and Glideslope tracking
1	27” Industry Standard	All	Flat Panel displays for exterior views and virtual instruments
1	CH Products Corporation	All	Pro Pedals (Rudder Pedals)
1	SolidTECK	All	Keyboard and track pad
1	Redbird Flight Simulations	All	Yoke Control
1	Redbird Flight Simulations	All	Throttle Quadrant
1	Redbird Flight Simulations	All—See Conf	Virtual Instrument Controls Overlay
1	Redbird Flight Simulations	All	Lower Panel Switch Block

Approved Individual Configurations Include:

TD BASIC-Standard: Approved Configuration

Yoke: Standard yoke. See Figure— 1
Throttle Quadrant: Single Engine Vernier. See Figure— 2
Virtual Instrument Controls Overlay: Traditional Round Gauges, Redbird 530, Redbird 140 Autopilot. See Figure— 8, 10

TD BASIC-Glass: Approved Configuration

Yoke: Standard yoke. See Figure— 1
Throttle Quadrant: Single Vernier. See Figure— 2
Virtual Instrument Controls Overlay: Glass Cockpit Redbird 1000 PFD & MFD with Integrated Redbird 700 Autopilot. See Figure— 9, 10

Section 3: Statements of Compliance

All Configurations, as noted, meet AC 61-136, Appendix 2, Paragraph 8 BATD Design Criteria:

a. Controls:

(1) Meets Requirements: Physical flight and aircraft system controls are recognizable as to their function and how they are manipulated solely from their appearance. The keyboard and track pad device supplied with the Redbird TD is used for setting flight location and flight conditions before a flight simulation session begins. During the simulation session the keyboard is used to display the pilot's performance, show backup instruments, introduce aircraft system failures and weather events, pause and end the flight. See Figures— 1, 2, 3, 4, 10

(2) Meets Requirements: Virtual controls are limited to the initialization of the device and to control aircraft configuration, location and weather. Avionics are virtually displayed but are operated with physical buttons and knobs that closely represent the configuration of the aircraft. See Figures— 7

(3) Meets Requirements: The keyboard and track pad is not used to set or position any feature of the represented aircraft for the maneuvers or flight training to be accomplished.

(4) Meets Requirements: The arrangement, appearance, and operation of controls, instruments, and switches closely represent the aircraft, including: master/battery, magnetos, alternators, fuel boost pump, avionics master, pitot heat, and required lights. See Figures— 3, 4

(5) Meets Requirements: During start-up of the device the PC configuration and file inventory is examined to ensure only tested software is installed. In the event additional files are found, an initialization error will be displayed and starting of a flight simulation session is not permitted.

b. Control Requirements:

(1)(a): Meets Requirements: A self-centering displacement yoke that allows continuous adjustment of pitch and bank is provided. See Figure— 1

(1)(b): Meets Requirements: Self-centering rudder pedals that allow continuous adjustment of yaw is provided.

(1)(c): Meets Requirements: Throttle control that allows continuous movement from idle to full power is provided. See Figure— 2

(1)(d): Meets Requirements: Mixture and throttle controls as applicable to the family of aircraft represented. See Figure— 2

(1)(e): Meets Requirements: See individual configuration.

c. Control Inputs:

(1) Meets Requirements: The transport delay for all control inputs is 300 milliseconds or less. The required transport delay is tested at system start-up by internal start/stop timers. Redbird Flight Simulations is the manufacturer of all control inputs and uses USB 2.0 which exceeds this requirement.

(2) Meets Requirements: Each control input is tested at training session start-up. If any device fails to meet the requirements of Appendix 2, Paragraph 8, Section c, item (1) a warning message is displayed and the device will not start a training session.

d. Display Requirements:

(1) Meets Requirements: See individual configuration.
(2) Meets Requirements: The display is updated at rate that insures that the image of the instrument does not appear to: be out of focus, illegible, jump, step, jagged, nor is it distracting.
(3) Meets Requirements: Display updates exceed 60Hz. All replicated instruments display changes equal or greater than the values required by this item.
(4) Meets Requirements: Displays reflect the dynamic behavior of an actual aircraft in flight.

e. Flight Dynamics Requirements:

(1) Meets Requirements: All simulated aircraft flight dynamics are comparable to the way an actual aircraft performs and handles.
(2) Meets Requirements: All simulated aircraft performance parameters are comparable to the aircraft being represented.
(3) Meets Requirements: Vertical lift component changes as a function of bank, comparable to the aircraft being represented.
(4) Meets Requirements: Changes in aircraft configuration are accompanied by changes in flight dynamics, comparable to the aircraft being represented.
(5) Meets Requirements: The handling and performance qualities of the simulated aircraft are affected by the presence and intensity of wind and turbulence in manner comparable to the aircraft being represented.

f. Instructional Management Requirements:

(1) Meets Requirements: The instructor has the ability to pause the system at any time using the supplied keyboard. See Figure— 5, 6
(2) Meets Requirements: The instructor as the ability to manipulate the parameters listed in this item independently of the simulation.
(3) Meets Requirements: The system records both the horizontal and vertical track of the aircraft movement and this can be view in real-time or at a later time via the Analyze Flight key on the supplied keyboard.
(4) Meets Requirements: The instructor is able to disable instruments prior to or during a training session and simulate failure of the instruments without stopping or freezing the simulation to affect the failure.
(5) Meets Requirements: The device includes a navigational data base that covers all of North America and is based on procedures as published per 14 CFR part 97.

Approved Configuration: TD BASIC- Standard

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with traditional "round" gauge instruments, Redbird 530, and Redbird 140 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 530).
Clock and timer.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Gear Handle—N/A by type of aircraft represented.
Carburetor Heat—N/A by type of aircraft represented.
Cowl Flaps—N/A by type of aircraft represented.

See Figures— 2, 3, 4, 8

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:

Traditional flight instruments. See Figure— 8
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 530)
Navigation radio that meets the requirements listed in this item. (Redbird 530)

Approved Configuration: TD BASIC- Glass

This configuration represents a single engine land, fixed gear, fixed pitch propeller, high wing, airplane with a Glass Cockpit Redbird 1000 PFD & MFD and an Integrated Redbird 700 Autopilot.

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria
b. Control Requirements:

(1)(e) This configuration has controls for:
Wing Flaps.
Pitch Trim.
Communication and Navigation Radios (Redbird 1000).
Clock and timer.
Transponder.
Altimeter.
Microphone with push-to-talk switch.

This configuration does not have controls for:
Gear Handle—N/A by type of aircraft represented.
Carburetor heat—N/A by type of aircraft represented.
Cowl flaps—N/A by type of aircraft represented.
See Figures— 2, 3, 4, 9

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the aircraft being represented:
Glass Cockpit Redbird 1000 PFD & MFD with standby instruments. See Figure— 9
Sensitive altimeter that meets the requirements listed in this item.
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
Airspeed indicator that meets the requirements listed in this item.
Vertical speed indicator that meets the requirements listed in this item.
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
Slip and skid indicator that meets the requirements listed in this item.
Engine instruments applicable to the aircraft being represented.
Suction gauge is not applicable to the aircraft being represented.
Flap setting indicator applicable to the aircraft being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird 1000)
Navigation radio that meets the requirements listed in this item. (Redbird 1000)

Section 4: Figures and Photos

Figure 1: Yoke- Standard



Figure 2: Throttle Quadrant- Single Engine Vernier



Figure 3: Switch Panel- Left Side

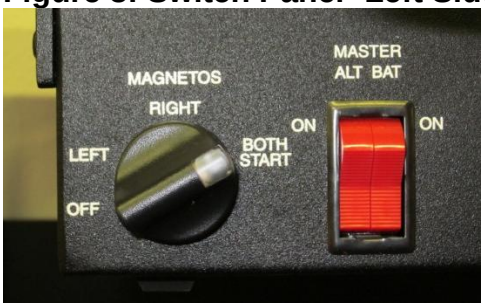


Figure 4: Switch Panel- Right Side



Figure 5: Keyboard and Trackpad



Figure 6: Instructor's Keys



Figure 7: Flight Simulation Session Start Screen

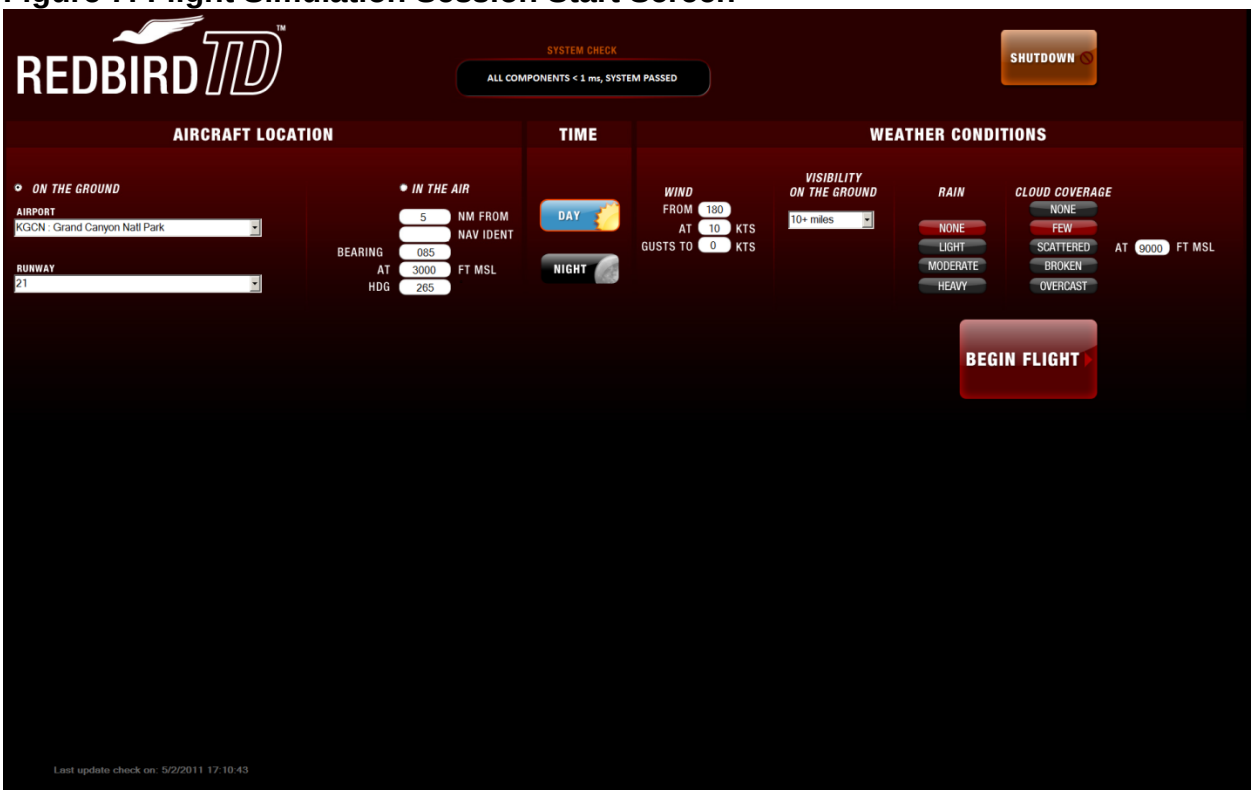


Figure 8: TD BASIC- Standard Instrument Panel



Figure 9: TD BASIC-Glass Instrument Panel



Figure 10: Assembled Unit

